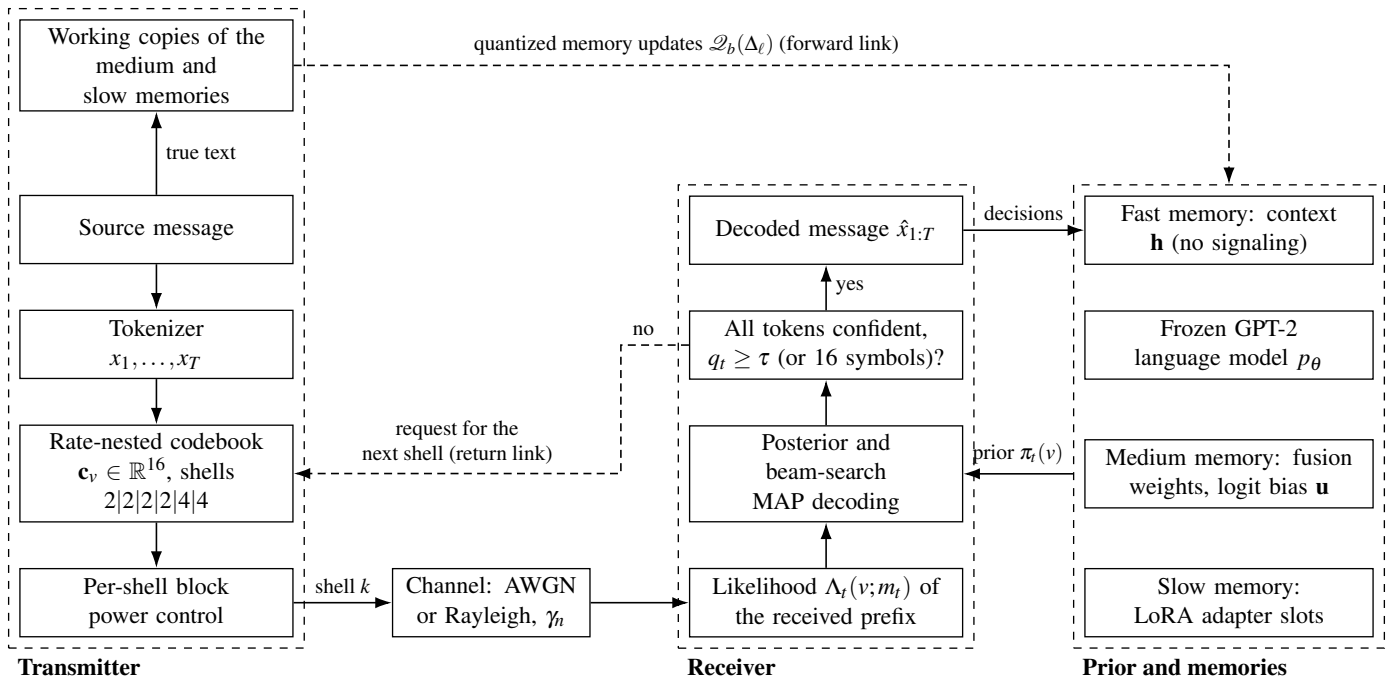

Working copies of the medium and slow memories
quantized memory updates $\mathscr{Q}_b(\Delta_\ell)$ (forward link)
true text
Source message
Tokenizer $x_1, \ldots, x_T$
Rate-nested codebook $\mathbf{c}_v \in \mathbb{R}^{16}$, shells 2|2|2|2|4|4
Per-shell block power control
shell k
Channel: AWGN or Rayleigh, γ_n
Likelihood $\Lambda_t(v; m_t)$ of the received prefix
Posterior and beam-search MAP decoding
All tokens confident, $q_t \geq \tau$ (or 16 symbols)?
no
request for the next shell (return link)
yes
Decoded message $\hat{x}_{1:T}$
decisions
prior $\pi_t(v)$
Fast memory: context $\mathbf{h}$ (no signaling)
Frozen GPT-2 language model p_θ
Medium memory: fusion weights, logit bias $\mathbf{u}$
Slow memory: LoRA adapter slots
Transmitter
Receiver
Prior and memories